الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

GRADE 10 ADVANCED · MATHEMATICS

End of Term 1
Revision Booklet

UNIT 2
Quadrilaterals

UNIT 3
Similarity

مراجعة نهاية الفصل الأول وفق الهيكل · الوحدتان الثانية والثالثة

12	96	12
SECTIONS	QUESTIONS	TEXTBOOK PANELS

Student _____

Class _____

Al Khalidia Boys' School · Cycle 3 · Teacher: Mr. Yousef Al Yousef · 2026–2027

بالميثاق نبلغ الآفاق

Contents

Every section follows one coverage question of the End of Term 1 structure, in the order of the lessons. Each one gives the key idea, the method, a worked example, the textbook's own exercises and eight practice questions. The answers are at the back.

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How to use this booklet

1. Read the **Key idea** and the **Method** first — out loud if you can.
2. Cover the **Worked example** and try it yourself, then compare line by line.
3. Solve the **textbook exercises** in your notebook: they are the coverage questions themselves.
4. Do the **Practice**, then mark it with the **Answer key** and read the reason for every slip.

UNIT 2

Quadrilaterals

الأشكال الرباعية

2-1 Angles of Polygons

2-2 Parallelograms

2-3 Tests for Parallelograms

2-6 Trapezoids and Kites

Section 1 | Coverage Q5 | Unit 2 · Lesson 2-1

Angles of Polygons

Find and use the sum of the measures of the interior angles of a polygon

إيجاد مجموع قياسات الزوايا الداخلية لمضلّع واستخدامه

Textbook · Exercises 1–4 · p. 63

KEY IDEA

The interior angles of a convex n**-sided polygon** add up to $(n-2)\cdot 180°$. In a **regular** polygon every angle is equal: divide the sum by n.

مجموع الزوايا الداخلية لمضلّع محدّب عدد أضلاعه n يساوي $(n-2)\cdot 180°$. في المضلّع **المنتظم** كل الزوايا متساوية: اقسم المجموع على n.

MUST KNOW

- Sum $= (n-2)\cdot 180°$.
- Regular polygon: each angle $= \dfrac{(n-2)\cdot 180°}{n}$.
- Write an equation by adding all the angle expressions.
- A quadrilateral: $360°$; a pentagon: $540°$; a hexagon: $720°$.

METHOD

1. Count the sides n.
2. Find the sum $(n-2)\cdot 180°$.
3. For algebra angles, set their sum equal to it and solve for x.
4. Substitute x back to find each angle.

WORKED EXAMPLE

Ex. 5 (P63): each angle of a regular octagon.

1. $n = 8$: sum $= (8-2)\cdot 180° = 1080°$.
2. Each angle $= 1080° \div 8$.
3. $= 135°$.

WATCH OUT

- Using $n \cdot 180°$ instead of $(n-2)\cdot 180°$.
- Forgetting to divide by n for a regular polygon.
- Setting the sum of the angle expressions equal to $180°$.
- Finding x but not the angle.

CHECK YOUR ANSWER

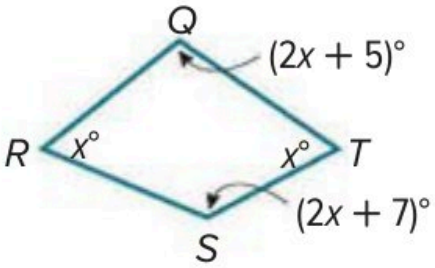
✓ All angles add up to the sum.

✓ Each angle is less than $180°$ in a convex polygon.

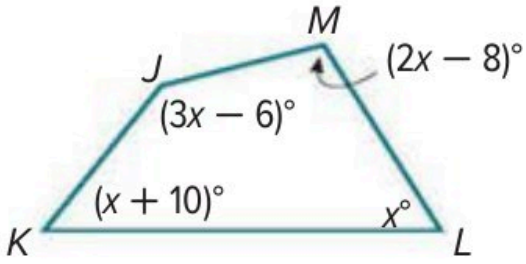
Example 1

Find the measure of each interior angle.

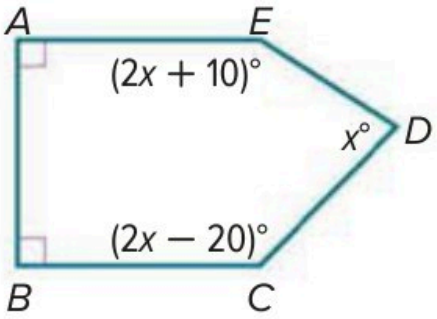
1.



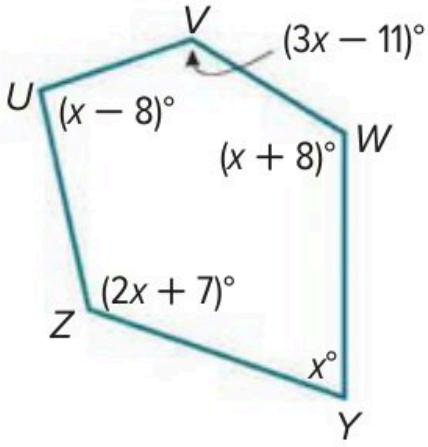
2.



3.



4.



Practice

Choose the correct answer. Show your working in the space beside each question.

1.1 **Find the sum of the interior angles.**

A convex hexagon.

(A) 540° (B) 720°

(C) 1080° (D) 360°

1.2 **Find the measure of each interior angle.**

A regular octagon (the Tribune room, Ex. 5).

(A) 135° (B) 1080°

(C) 45° (D) 108°

1.3 **Find the largest interior angle.**

Quadrilateral $QRST$ has $m\angle Q = (2x + 5)^\circ$, $m\angle R = x^\circ$, $m\angle T = x^\circ$ and $m\angle S = (2x + 7)^\circ$.

- (A) 58°
- (B) 121°
- (C) 123°
- (D) 126°

1.4 **Find the measure of $\angle E$.**

Pentagon $ABCDE$ has right angles at A and B, $m\angle C = (2x - 20)^\circ$, $m\angle D = x^\circ$ and $m\angle E = (2x + 10)^\circ$.

- (A) 74°
- (B) 158°
- (C) 128°
- (D) 148°

1.5 **Find the number of sides.**

Each interior angle of a regular polygon measures 144°.

- (A) 8
- (B) 12
- (C) 9
- (D) 10

1.6 **Find the number of sides.**

Each interior angle of a regular polygon measures 160°.

- (A) 9
- (B) 18
- (C) 20
- (D) 16

1.7 **Find the measure of each interior angle.**

A regular heptagon (an animal pen). Round to the nearest tenth.

- (A) 900°
- (B) 140°
- (C) 135°
- (D) 128.6°

1.8 **Find the measure of each interior angle.**

A regular dodecagon (12 sides).

- (A) 150°
- (B) 1800°
- (C) 30°
- (D) 144°

Section 2 | Coverage Q22 | Unit 2 · Lesson 2-1

Angles of Polygons

Find and use the sum of the measures of the exterior angles of a polygon

إيجاد مجموع قياسات الزوايا الخارجية لمضلّع واستخدامه

Textbook · Exercises 9–18 · p. 64

KEY IDEA

The exterior angles of any convex polygon (one at each vertex) add up to $360°$. In a regular n-gon each exterior angle is $\frac{360°}{n}$, and interior + exterior $= 180°$.

مجموع الزوايا الخارجية لأي مضلّع محدّب (واحدة عند كل رأس) يساوي $360°$. في المضلّع المنتظم ذي n ضلعًا كل زاوية خارجية = $\frac{360°}{n}$، ومجموع الداخلية والخارجية $= 180°$.

MUST KNOW

- Sum of exterior angles $= 360°$.
- Regular polygon: exterior $= \frac{360°}{n}$.
- Interior + exterior $= 180°$.
- Number of sides $= \frac{360°}{\text{exterior angle}}$.

METHOD

1. Identify the exterior angles.
2. Set their sum equal to $360°$ if they involve x, and solve.
3. For a regular polygon use $360° \div n$ or $n = 360° \div$ exterior.
4. Convert to the interior angle if needed.

WORKED EXAMPLE

Ex. 10 (P64): interior angle $156°$, find the number of sides.

1. Exterior $= 180° - 156° = 24°$.
2. $n = 360° \div 24°$.
3. $n = 15$.

WATCH OUT

- Using $(n - 2) \cdot 180°$ for exterior angles.
- Forgetting that the exterior angle is the supplement of the interior angle.
- Not getting a whole number of sides and not rechecking.
- Taking two exterior angles at one vertex.

CHECK YOUR ANSWER

✓ The exterior angles add up to $360°$.

✓ n is a whole number.

Example 3

The measure of an interior angle of a regular polygon is given. Find the number of sides in the polygon.

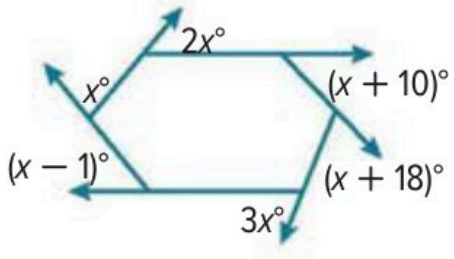
9. 144° **10.** 156° **11.** 160°

12. 108° **13.** 120° **14.** 150°

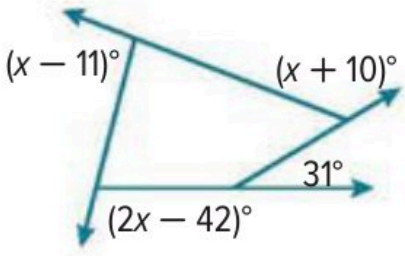
Example 4

Find the value of *x* in each diagram.

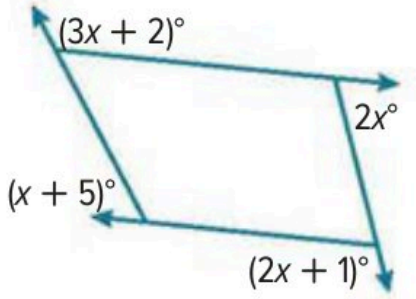
15.



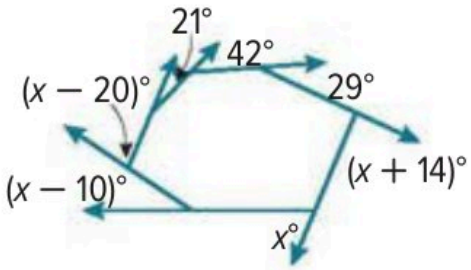
16.



17.



18.



Practice

Choose the correct answer. Show your working in the space beside each question.

2.1 **Find the sum of the exterior angles.**

A convex octagon (one exterior angle at each vertex).

(A) 1080° (B) 135°

(C) 720° (D) 360°

2.2 **Find the measure of each exterior angle.**

A regular decagon (10 sides).

- (A) $144°$
- (B) $36°$
- (C) $18°$
- (D) $72°$

2.3 **Find the number of sides.**

Each exterior angle of a regular polygon is $24°$.

- (A) 15
- (B) 12
- (C) 16
- (D) 18

2.4 **Find the number of sides.**

Each exterior angle of a regular polygon is $40°$.

- (A) 8
- (B) 10
- (C) 9
- (D) 12

2.5 **Find x.**

The exterior angles of a quadrilateral measure $(3x + 2)°$, $(2x)°$, $(2x + 1)°$ and $(x + 5)°$.

- (A) $x = 45$
- (B) $x = 44$
- (C) $x = 40$
- (D) $x = 88$

2.6 **Find the interior angle.**

The exterior angle of a regular polygon is $18°$. What is each interior angle?

- (A) $18°$
- (B) $20°$
- (C) $108°$
- (D) $162°$

2.7 **Find the number of sides.**

Each interior angle of a regular polygon is $108°$.

- (A) 4
- (B) 5
- (C) 6
- (D) 10

2.8 **Find x.**

The exterior angles of a quadrilateral measure $(x-11)^\circ$, $(x+10)^\circ$, 31° and $(2x-42)^\circ$.

(A) $x = 90$

(B) $x = 96$

(C) $x = 100$

(D) $x = 93$

Section 3 | Coverage Q6 | Unit 2 · Lesson 2-2

Parallelograms

Recognize and apply the properties of the sides and angles of parallelograms

التعرّف إلى خصائص أضلاع متوازي الأضلاع وزواياه وتطبيقها

Textbook · Exercises 9–12 · p. 72

KEY IDEA

In a parallelogram: **opposite sides** are parallel and congruent; **opposite angles** are congruent; **consecutive angles** are supplementary; the **diagonals bisect each other** and each separates it into two congruent triangles.

في متوازي الأضلاع: **الضلعان المتقابلان** متوازيان ومتطابقان؛ **الزاويتان المتقابلتان** متطابقتان؛ **الزاويتان المتتاليتان** متكاملتان (مجموعهما 180°)؛ و**القطران ينصّف كل منهما الآخر**.

MUST KNOW

- Opposite sides: equal.
- Opposite angles: equal.
- Consecutive angles: sum $180°$.
- Diagonals bisect each other (but are not equal in general).

METHOD

1. Mark the equal opposite sides and angles.
2. Write an equation from the property you need.
3. Solve for the variable.
4. Substitute back to find the required measure.

WORKED EXAMPLE

Find b if opposite sides are $3b + 5$ and $b + 11$.

1. Opposite sides are equal: $3b + 5 = b + 11$.
2. $2b = 6 \Rightarrow b = 3$.
3. Each side $= 3(3) + 5 = 14$.

WATCH OUT

- Using equal consecutive angles instead of supplementary.
- Thinking the diagonals are equal.
- Solving for x and stopping before finding the measure.
- Mixing up which sides are opposite.

CHECK YOUR ANSWER

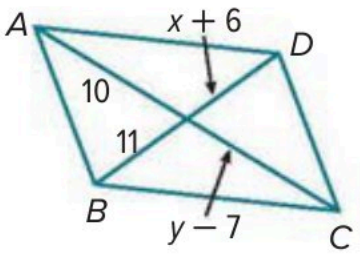
✓ Opposite sides come out equal.

✓ Consecutive angles add to $180°$.

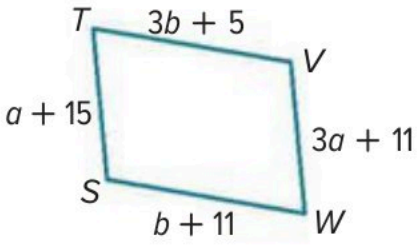
Example 3

Find the value of each variable in each parallelogram.

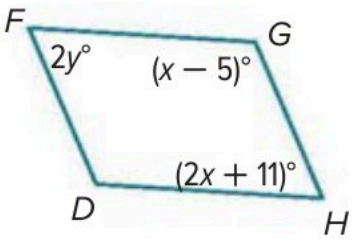
9.



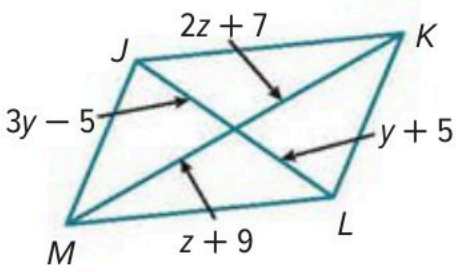
10.



11.



12.



Practice

Choose the correct answer. Show your working in the space beside each question.

3.1 **Find the measure of an angle.**

In parallelogram $PQRS$, $m\angle P = 128°$. What is $m\angle S$?

(A) $52°$ (B) $128°$

(C) $62°$ (D) $232°$

3.2 **Find the measure of a side.**

In parallelogram $PQRS$, $PQ = 3$ and $QR = 5$. What is SR?

(A) 5 (B) 8

(C) 4 (D) 3

3.3 **Find the value of b.**

In parallelogram $TVWS$ the opposite sides $\overline{TV}$ and $\overline{SW}$ measure $3b + 5$ and $b + 11$.

(A) $b = 6$ (B) $b = 3$

(C) $b = 4$ (D) $b = -3$

3.4 **Find the value of x.**

In parallelogram $FGHD$, the consecutive angles $\angle G$ and $\angle H$ measure $(x-5)^\circ$ and $(2x+11)^\circ$.

(A) $x = 58$

(B) $x = 62$

(C) $x = 41$

(D) $x = 87$

3.5 **Find the length of a diagonal.**

The diagonals of parallelogram $JKLM$ meet at P, with $JP = 3y - 5$ and $PL = y + 5$. Find JL.

(A) 10

(B) 15

(C) 20

(D) 5

3.6 **Find the area.**

A dog park is a parallelogram with vertices $(-8, 0)$, $(-4, -5)$, $(9, 0)$, $(5, 5)$, and 1 unit = 5 m. What is its area?

(A) $85\ \text{m}^2$

(B) $2125\ \text{m}^2$

(C) $425\ \text{m}^2$

(D) $1700\ \text{m}^2$

3.7 **Which statement is not true for every parallelogram?**

(A) Opposite sides are congruent.

(B) Opposite angles are congruent.

(C) The diagonals bisect each other.

(D) The diagonals are congruent.

3.8 **Find the other angles.**

One angle of a parallelogram measures 70°. What are all four angle measures, in order?

(A) $70^\circ, 70^\circ, 110^\circ, 110^\circ$

(B) $70^\circ, 110^\circ, 70^\circ, 110^\circ$

(C) $70^\circ, 90^\circ, 70^\circ, 90^\circ$

(D) $70^\circ, 110^\circ, 110^\circ, 70^\circ$

Section 4 | Coverage Q7 | Unit 2 · Lesson 2-3

Tests for Parallelograms

Recognize the conditions that ensure a quadrilateral is a parallelogram

التعرّف إلى الشروط التي تضمن أن الشكل الرباعي متوازي أضلاع

Textbook · Learn · p. 75 · Exercises 1–6 · p. 79

KEY IDEA

A quadrilateral is a parallelogram if: **both pairs of opposite sides are congruent**; or **both pairs of opposite angles are congruent**; or its **diagonals bisect each other**; or **one pair of opposite sides is both parallel and congruent**.

الشكل الرباعي متوازي أضلاع إذا: **تطابق كل ضلعين متقابلين**، أو **تطابقت كل زاويتين متقابلتين**، أو **نصّف قطراه بعضهما**، أو **وُجد ضلعان متقابلان متوازيان ومتطابقان**.

MUST KNOW

- Four tests: two pairs of opposite sides, two pairs of opposite angles, diagonals bisecting, one pair parallel and congruent.
- One congruent pair alone is not enough.
- To make a quadrilateral a parallelogram, set the opposite sides (or angles) equal.
- Say which theorem you used.

METHOD

1. Read what is given.
2. Match it to one of the four tests.
3. If x, y are asked, set the opposite measures equal and solve.
4. Write the conclusion with the reason.

WORKED EXAMPLE

Ex. 7 (P79): opposite sides $6x + 1,\ 8x - 3$ and $4y - 7,\ 2y + 3$.

1. $6x + 1 = 8x - 3 \Rightarrow x = 2$.
2. $4y - 7 = 2y + 3 \Rightarrow y = 5$.
3. Both pairs of opposite sides are then equal.

WATCH OUT

- Accepting one pair of congruent sides as proof.
- Mixing opposite and consecutive sides.
- Setting consecutive angles equal.
- Giving a conclusion without a reason.

CHECK YOUR ANSWER

✓ The test you used needs **both** pairs (or the parallel-and-congruent pair).

✓ Both pairs of opposite sides come out equal.

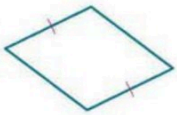
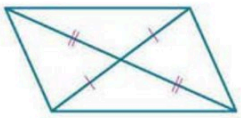
Learn Tests for Parallelograms

If a quadrilateral has each pair of opposite sides parallel, it is a parallelogram by definition. This is not the only test, however, that can be used to determine if a quadrilateral is a parallelogram.

Theorems: Conditions for Parallelograms
Theorem 2.9
If both pairs of opposite sides of a quadrilateral are congruent, then the quadrilateral is a parallelogram.
Theorem 2.10
If both pairs of opposite angles of a quadrilateral are congruent, then the quadrilateral is a parallelogram.
Theorem 2.11
If the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.
Theorem 2.12
If one pair of opposite sides of a quadrilateral is both parallel and congruent, then the quadrilateral is a parallelogram.

Example 1

Determine whether each quadrilateral is a parallelogram. Justify your answer.

1. 
2. 
3. 
4. 
5. 
6. 

Practice

Choose the correct answer. Show your working in the space beside each question.

4.1 **Is the quadrilateral a parallelogram?**

A quadrilateral has sides 12 cm, 5 cm, 12 cm, 5 cm in order.

- (A) No: only one pair of sides is congruent
- (B) Yes: the diagonals are congruent
- (C) Yes: both pairs of opposite sides are congruent
- (D) Not enough information

4.2 **Is the quadrilateral a parallelogram?**

Its angles, in order, measure $98°$, $82°$, $98°$, $82°$.

- (A) Yes: both pairs of opposite angles are congruent
- (B) No: the angles are not all equal
- (C) Yes: all the angles are supplementary
- (D) Not enough information

4.3 **Is the quadrilateral a parallelogram?**

The diagonals AC and BD meet at E with $AE = EC = 7$ and $BE = ED = 5$.

- (A) No: the diagonals have different lengths
- (B) Yes: the diagonals are perpendicular
- (C) Not enough information
- (D) Yes: the diagonals bisect each other

4.4 **Is the quadrilateral a parallelogram?**

Only one fact is known: $AB = CD = 6$.

- (A) Yes: a pair of opposite sides is congruent
- (B) Not enough information
- (C) Yes: by the definition
- (D) No: all four sides must be equal

4.5 **Find x and y.**

A quadrilateral has opposite sides $6x + 1$ and $8x - 3$, and $4y - 7$ and $2y + 3$. For which values is it a parallelogram?

(A) $x = 2,\ y = 5$

(B) $x = 2,\ y = 2$

(C) $x = 5,\ y = 2$

(D) $x = -2,\ y = 5$

4.6 **Find x and y.**

Opposite angles measure $(5x + 29)°$ and $(7x - 11)°$, and $(5y - 9)°$ and $(3y + 15)°$.

(A) $x = 12,\ y = 20$

(B) $x = 20,\ y = 24$

(C) $x = 20,\ y = 12$

(D) $x = 10,\ y = 12$

4.7 **Find x and y.**

The diagonals of a quadrilateral cross at a point. One diagonal is cut into $-4x - 2$ and $-3x + 4$; the other into $3y - 5$ and $2y + 18$. For which values is it a parallelogram?

(A) $x = 6,\ y = 23$

(B) $x = -6,\ y = 23$

(C) $x = -6,\ y = 13$

(D) $x = -2,\ y = 23$

4.8 **Is it a parallelogram?**

$A(0,0),\ B(4,1),\ C(7,5),\ D(3,4)$

(A) No: $AB \nparallel BC$

(B) No: the diagonals are not equal

(C) Not enough information

(D) Yes: $\overrightarrow{AB} = \overrightarrow{DC} = (4,1)$ and $\overrightarrow{AD} = \overrightarrow{BC} = (3,4)$

Section 5 | Coverage Q8

Unit 2 · Lesson 2-6

Trapezoids and Kites

Recognize and apply the properties of kites

التعرّف إلى خصائص الطائرة الورقية وتطبيقها

Textbook · Exercises 13–16 · p. 106

KEY IDEA

A **kite** has two pairs of adjacent congruent sides; its **diagonals are perpendicular** and exactly one pair of opposite angles (between the unequal sides) is congruent. Midsegment of a trapezoid $= \frac{1}{2}(b_1 + b_2)$.

الطائرة الورقية لها زوجان من الأضلاع المتجاورة المتطابقة؛ **قطراها متعامدان** وفيها زاويتان متقابلتان متطابقتان فقط. القطعة الواصلة بين منتصفي ساقي شبه المنحرف $= \frac{1}{2}(b_1 + b_2)$.

MUST KNOW

- Diagonals of a kite are perpendicular: use right triangles.
- Angles of any quadrilateral add to $360°$.
- Isosceles trapezoid: equal legs, equal base angles, equal diagonals.
- Midsegment $= \frac{b_1+b_2}{2}$.

METHOD

1. Mark the equal sides/angles.
2. Use the right triangles made by the perpendicular diagonals.
3. Use $360°$ for angle problems.
4. Round only at the end.

WORKED EXAMPLE

Kite with $AE = 4$, $EC = 7$, $BE = 6$, diagonals perpendicular: find BC.

1. $\triangle BEC$ is right at E.
2. $BC = \sqrt{7^2 + 6^2} = \sqrt{85}$.
3. It does not simplify further.

WATCH OUT

- Treating the kite like a parallelogram (opposite sides equal).
- Forgetting that the diagonals are perpendicular.
- Splitting $360°$ equally among all four angles.
- Leaving $\sqrt{52}$ unsimplified when asked for simplest form.

CHECK YOUR ANSWER

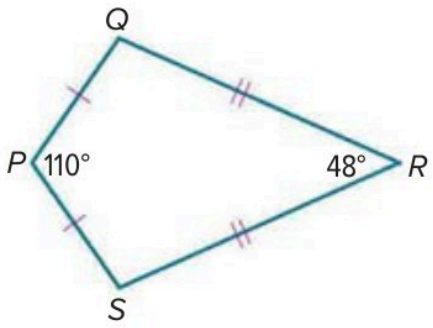
✓ The four angles add to $360°$.

✓ Radicals are in simplest form.

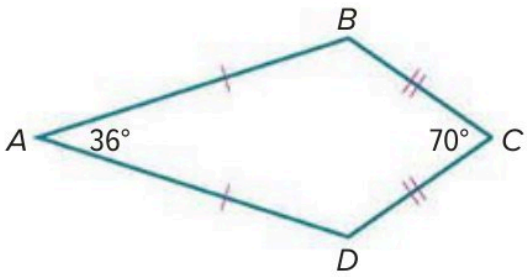
Example 6

Find each measure in the kites.

13. $m\angle Q$



14. $m\angle D$

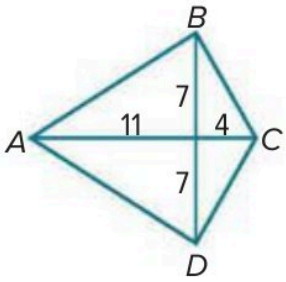


Example 7

15. REASONING Quadrilateral *ABCD* is a kite.

a. Find *BC*. Write your answer in simplest radical form.

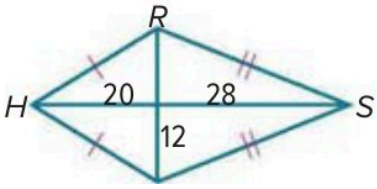
b. Find the perimeter of kite *ABCD*. Round your answer to the nearest tenth, if necessary.



16. REASONING Quadrilateral *HRSE* is a kite.

a. Find *RH*. Write your answer in simplest radical form.

b. Find the perimeter of kite *HRSE*. Round your answer to the nearest tenth, if necessary.



Practice

Choose the correct answer. Show your working in the space beside each question.

5.1 **Find the length of a side.**

Kite $ABCD$: the diagonals meet at E with $AE = 11$, $EC = 4$ and $BE = ED = 7$, and $AC \perp BD$. Find BC in simplest radical form.

(A) $\sqrt{170}$ (B) $\sqrt{130}$

(C) $\sqrt{65}$ (D) 11

5.2 **Find the perimeter.**

The same kite $ABCD$: $AB = AD = \sqrt{170}$ and $CB = CD = \sqrt{65}$. Round to the nearest tenth.

(A) 21.1 (B) 38.0

(C) 42.2 (D) 27.0

5.3 **Find the measure of an angle.**

In kite $PQRS$, $\angle Q \cong \angle S$, $m\angle P = 110°$ and $m\angle R = 48°$. Find $m\angle Q$.

(A) $101°$ (B) $92°$

(C) $79°$ (D) $202°$

5.4 **Find the measure of an angle.**

In kite $ABCD$, $\angle B \cong \angle D$, $m\angle A = 36°$ and $m\angle C = 70°$. Find $m\angle D$.

(A) $254°$ (B) $107°$

(C) $137°$ (D) $127°$

5.5 **The diagonals of a kite are**

(A) congruent (B) perpendicular

(C) parallel (D) bisecting each other

5.6 **Find x so that the trapezoid is isosceles.**

In trapezoid $ABCD$, $AC = 3x - 7$ and $BD = 2x + 8$.

(A) $x = 15$ (B) $x = 1$

(C) $x = 3$ (D) $x = 5$

5.7 **Find x so that the trapezoid is isosceles.**

In trapezoid $ABCD$, $m\angle ABC = (4x + 11)°$ and $m\angle DAB = (2x + 33)°$.

(A) $x = 22$ (B) $x = 44$

(C) $x = 11$ (D) $x = 17$

5.8 **Find the midsegment.**

The bases of a trapezoid are 11 and 18. Find the length of the midsegment.

(A) 29 (B) 14.5

(C) 7 (D) 6.5

UNIT 3

Similarity

التشابه

3-1 Dilations

3-2 Similar Polygons

3-3 Similar Triangles: AA Similarity

3-4 Similar Triangles: SSS and SAS Similarity

3-5 Triangle Proportionality

3-6 Parts of Similar Triangles

Section 6 | Coverage Q9

Unit 3 · Lesson 3-1

Dilations

Find the scale factor of a dilation

إيجاد معامل القياس لتمدّد

Textbook · Exercises 13–16, 20 · p. 120–121

KEY IDEA

A dilation with scale factor k and center the origin sends (x, y) to (kx, ky). The scale factor is $k = \frac{\text{image}}{\text{original}}$. $k > 1$: enlargement; $0 < k < 1$: reduction.

التمدّد بمعامل k ومركزه نقطة الأصل يحوّل (x, y) إلى (kx, ky). معامل القياس $k = \frac{\text{الصورة}}{\text{الأصل}}$. إذا $k > 1$ تكبير، وإذا $0 < k < 1$ تصغير.

MUST KNOW

- Image $= k\times$ original (multiply every coordinate).
- $k = \frac{\text{image length}}{\text{original length}}$ — image on top.
- $k > 1$ enlarges; $0 < k < 1$ shrinks.
- Always check that every vertex gives the same k.

METHOD

1. Pick a pair of matching points or lengths.
2. Divide image by original.
3. Check with a second pair.
4. State enlargement or reduction.

WORKED EXAMPLE

Ex. 20 (P121): $C(5, 2)$, $k = 2.5$.

1. $C' = (2.5 \cdot 5,\ 2.5 \cdot 2)$.
2. $= (12.5,\ 5)$.

WATCH OUT

- Dividing original by image.
- Adding k to the coordinates instead of multiplying.
- Using only one coordinate.
- Calling $k = 0.4$ an enlargement.

CHECK YOUR ANSWER

✓ The same k works for every vertex.

✓ Image over original, not the opposite.

Mixed Exercises

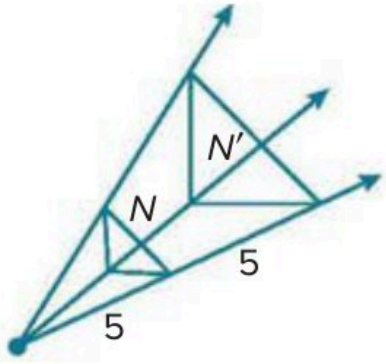
Graph the image of each polygon with the given vertices after a dilation centered at the origin with the given scale factor.

13. $F(-10, 4)$, $G(-4, 4)$, $H(-4, -8)$, $k = 0.25$ **14.** $X(2, -1)$, $Y(-6, 4)$, $Z(-2, -5)$, $k = \frac{5}{4}$

15. $M(4, 6)$, $N(-6, 2)$, $P(0, -8)$, $k = \frac{3}{4}$

16. $R(-2, 6)$, $S(0, -1)$, $T(-5, 3)$, $k = 1.5$

19. Determine whether the dilation from Figure N to N' is an *enlargement* or a *reduction*. Find the scale factor of the dilation.



20. $\triangle ABC$ has vertices $A(2, 2)$, $B(3, 4)$, and $C(5, 2)$. What are the coordinates of point C of the image of the triangle after a dilation centered at the origin with a scale factor of 2.5?

Practice

Choose the correct answer. Show your working in the space beside each question.

6.1 **Find the image of a vertex.**

Dilate $F(-10, 4)$ with center the origin and scale factor $k = 0.25$.

(A) $F'(-2.5, 1)$

(B) $F'(-40, 16)$

(C) $F'(-2.5, -1)$

(D) $F'(-9.75, 3.75)$

6.2 **Find the image of a vertex.**

Dilate $X(2, -1)$ with center the origin and scale factor $k = \frac{5}{4}$.

(A) $X'\left(\frac{8}{5}, -\frac{4}{5}\right)$

(B) $X'\left(\frac{13}{4}, \frac{1}{4}\right)$

(C) $X'\left(\frac{5}{2}, -\frac{5}{4}\right)$

(D) $X'\left(\frac{5}{2}, \frac{5}{4}\right)$

6.3 **Find the image of a vertex.**

Dilate $P(0, -8)$ with center the origin and scale factor $k = \frac{3}{4}$.

- (A) $P'(0,\ -32)$
- (B) $P'(\frac{3}{4},\ -8)$
- (C) $P'(0,\ -6)$
- (D) $P'(0,\ -10\frac{2}{3})$

6.4 **Find the image of a vertex.**

Dilate $R(-2, 6)$ with center the origin and scale factor $k = 1.5$.

- (A) $R'(-3,\ 9)$
- (B) $R'(-0.5,\ 1.5)$
- (C) $R'(-3,\ 7.5)$
- (D) $R'(2,\ 9)$

6.5 **Find the image of a vertex.**

$\triangle ABC$ has vertices $A(2, 2),\ B(3, 4),\ C(5, 2)$. Find C' after a dilation centered at the origin with scale factor 2.5.

- (A) $C'(7.5,\ 4.5)$
- (B) $C'(2,\ 0.8)$
- (C) $C'(12.5,\ 2)$
- (D) $C'(12.5,\ 5)$

6.6 **Enlargement or reduction?**

A dilation has scale factor $k = 0.4$. The image is

- (A) an enlargement (larger than the original)
- (B) a reduction (smaller than the original)
- (C) congruent to the original
- (D) a reflection of the original

6.7 **Find the scale factor.**

A dilation about the origin maps $A(2, 3)$ to $A'(6, 9)$. What is the scale factor?

- (A) $k = 3$
- (B) $k = \frac{1}{3}$
- (C) $k = 4$
- (D) $k = 6$

6.8 **Find the scale factor.**

A side of the original figure is 4 units long and the matching side of the image is 10 units. What is the scale factor of the dilation?

- (A) $k = 0.4$
- (B) $k = 6$
- (C) $k = 2.5$
- (D) $k = 40$

Section 7 | Coverage Q11

Unit 3 · Lesson 3-1

Dilations

Find the scale factor of a dilation

إيجاد معامل القياس لتمدّد

Textbook · Exercises 10–12, 17–18 · p. 120

KEY IDEA

Scale factor $= \dfrac{\text{image}}{\text{original}}$, measured from corresponding sides or coordinates. Two dilations with the same center multiply their scale factors.

معامل القياس $= \dfrac{\text{الصورة}}{\text{الأصل}}$ من أطوال أو إحداثيات متناظرة. تمدّدان لهما المركز نفسه معاملهما يتضاعف: $k_1 \cdot k_2$.

MUST KNOW

- Compare corresponding parts only.
- Image on top, original on the bottom.
- Check a second pair gives the same value.
- Scale factors of successive dilations multiply.

METHOD

1. Find a pair of corresponding sides or points.
2. Divide image by original.
3. Check another pair.
4. Classify: enlargement or reduction.

WORKED EXAMPLE

Ex. 12 (P120): logo 6 cm wide, image 15 cm wide.

1. $k = \dfrac{\text{image}}{\text{original}} = \dfrac{15}{6}$.
2. $k = 2.5$: an enlargement.

WATCH OUT

- Original over image.
- Comparing non-corresponding sides.
- Adding the two scale factors of successive dilations.
- Forgetting the negative sign when the image is on the other side of the center.

CHECK YOUR ANSWER

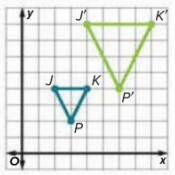
✓ Every pair gives the same k.

FROM YOUR TEXTBOOK Exercises 10–12, 17–18 · p. 120

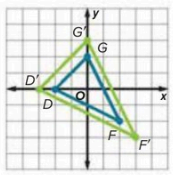
Example 4

Find the scale factor of the dilation.

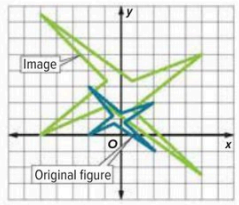
10. $\triangle J'K'P'$ is the image of $\triangle JKP$.



11. $\triangle D'F'G'$ is the image of $\triangle DFG$.

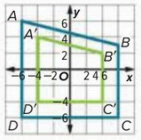


12. Tyrone drew a logo and a dilation of the same logo on the coordinate plane. What is the scale factor of the dilation?

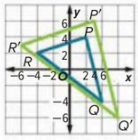


Find the scale factor of the dilation.

17. $A'B'C'D'$ is the image of $ABCD$.



18. $\triangle P'Q'R'$ is the image of $\triangle PQR$.



Practice

Choose the correct answer. Show your working in the space beside each question.

7.1 **Find the scale factor.**

$\triangle DFG$ has vertices $D(3,6),\ F(-6,3),\ G(0,-9)$. Its image has $D'(1,2),\ F'(-2,1),\ G'(0,-3)$. The scale factor is

(A) 3 (B) $-\frac{1}{3}$ (C) 2 (D) $\frac{1}{3}$

7.2 **Find the scale factor.**

A dilation maps $J(2,4),\ K(6,2)$ to $J'(5,10),\ K'(15,5)$. The scale factor is

(A) 2.5 (B) 0.4 (C) 3 (D) 13

7.3 **Find the scale factor.**

A logo is 6 cm wide. Its dilation is 15 cm wide. What is the scale factor?

(A) 0.4 (B) 9 (C) 2.5 (D) 90

7.4 **Find the scale factor.**

A segment $\overline{AB} = 8$ is dilated to $\overline{A'B'} = 6$. The scale factor is

(A) $\frac{4}{3}$ (B) 2 (C) $\frac{3}{4}$ (D) $\frac{7}{8}$

7.5 **Find the scale factor.**

A dilation about the origin maps the point $P(a, b)$ to $P'(-3a, -3b)$. The scale factor is

(A) $k = -3$

(B) $k = 3$

(C) $k = \frac{1}{3}$

(D) $k = -\frac{1}{3}$

7.6 **Combine two dilations.**

Davion enlarges a figure with scale factor 2, then dilates the result with scale factor $\frac{1}{3}$ (same center). One single dilation that maps the original to the final image has scale factor

(A) $\frac{3}{2}$

(B) $\frac{5}{3}$

(C) 6

(D) $\frac{2}{3}$

7.7 **Combine two dilations.**

A dilation with scale factor k is followed by a dilation with scale factor $\frac{1}{k}$ (same center). The final image is

(A) twice the original

(B) congruent to the original (in fact it is the original)

(C) half the original

(D) a reflection of the original

7.8 **Enlargement or reduction?**

Figure N is dilated to N′ and every length is multiplied by $\frac{5}{8}$. The dilation is

(A) a reduction with scale factor $\frac{5}{8}$

(B) an enlargement with scale factor $\frac{5}{8}$

(C) a reduction with scale factor $\frac{8}{5}$

(D) an enlargement with scale factor $\frac{8}{5}$

Section 8 | Coverage Q10

Unit 3 · Lesson 3-2

Similar Polygons

Solve problems using the properties of similar polygons

حلّ مسائل باستخدام خصائص المضلّعات المتشابهة

Textbook · Exercises 11–14 · p. 128

KEY IDEA

Similar polygons have **congruent corresponding angles** and **proportional corresponding sides**. Write the proportion with matching sides and solve by cross-multiplying.

المضلّعان المتشابهان زواياهما المتناظرة **متطابقة** وأضلاعهما المتناظرة **متناسبة**. نكتب النسبة بأضلاع متناظرة ثم نحلّها بالضرب التبادلي.

MUST KNOW

- Both conditions are needed: angles and sides.
- Match corresponding sides in the same order in each ratio.
- Cross-multiply to solve.
- Perimeters are in the same ratio k.

METHOD

1. List the corresponding sides.
2. Write a proportion with one unknown.
3. Cross-multiply and solve.
4. Check that the ratios agree.

WORKED EXAMPLE

Ex. 16 (P128): perimeter 126**, ratio** $5:2$**.**

1. Length $5k$, width $2k$.
2. $2(5k + 2k) = 126 \Rightarrow k = 9$.
3. Width $= 2 \cdot 9 = 18$ m.

WATCH OUT

- Writing the proportion with non-corresponding sides.
- Believing equal angles alone are enough for polygons with more than 3 sides.
- Adding instead of multiplying by the scale factor.
- Using the ratio of areas as the scale factor.

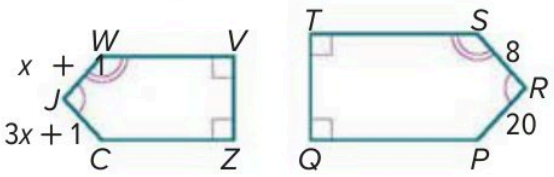
CHECK YOUR ANSWER

✓ Every pair of corresponding sides gives the same ratio.

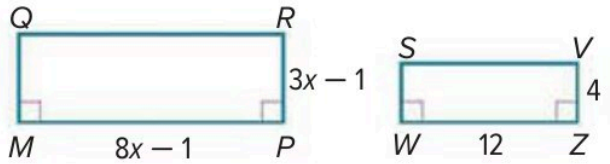
Example 3

Each pair of polygons is similar. Find the value of *x*.

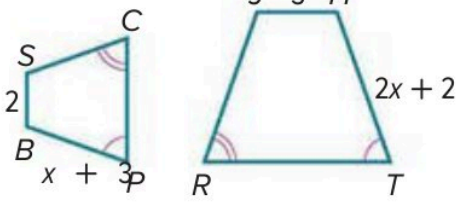
11.



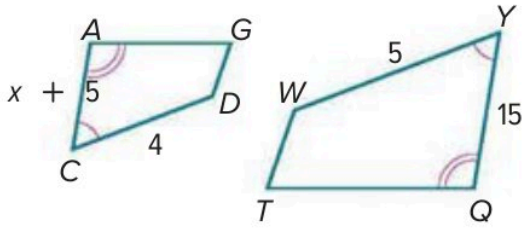
12.



13.



14.



Practice

Choose the correct answer. Show your working in the space beside each question.

8.1 **Find x.**

Two similar rectangles: the first is 8 by 12, the second is 6 by x.

- (A) $x = 16$
- (B) $x = 9$
- (C) $x = 4$
- (D) $x = 10$

8.2 **Find x.**

$ABCD \sim EFGH$ with $AB = x + 3$, $EF = 15$, $BC = 4$ and $FG = 10$.

- (A) $x = 6$
- (B) $x = 9$
- (C) $x = 1.5$
- (D) $x = 3$

8.3 **Find x.**

The polygons are similar: sides $2x + 2$ and 15 correspond, and sides 4 and 10 correspond.

- (A) $x = 2$
- (B) $x = 5.5$
- (C) $x = 3$
- (D) $x = 1$

8.4 **Find the width.**

Nayeli's rectangular lawn has perimeter 126 m and its length-to-width ratio is $5 : 2$. What is the width?

- (A) 36 m
- (B) 45 m
- (C) 18 m
- (D) 9 m

8.5 **Find the distance.**

On a map the home–store distance is 2 cm, store–dungeon 8 cm and dungeon–home 7 cm. The store–dungeon distance is really 4 km. How far is the whole trip?

- (A) 17 km
- (B) 4 km
- (C) 8.5 km
- (D) 34 km

8.6 **Are the rectangles similar?**

A 6×9 rectangle and an 8×12 rectangle. If they are similar, what is the scale factor from the **larger** to the **smaller**?

- (A) Yes, with scale factor $\frac{3}{4}$
- (B) No: the angles differ
- (C) No: $6 \not\equiv 8$
- (D) Yes, with scale factor $\frac{4}{3}$

8.7 **Are the rectangles similar?**

A 4×6 rectangle and a 6×8 rectangle.

- (A) Yes: all angles are $90°$
- (B) Yes: the scale factor is 1.5
- (C) Yes: they are both rectangles
- (D) No: the sides are not proportional

8.8 **Find the angle.**

$ABCD \sim WXYZ$ and $m\angle B = 75°$. What is $m\angle X$?

- (A) $105°$
- (B) $75°$
- (C) $15°$
- (D) $150°$

Section 9 | Coverage Q12 | Unit 3 · Lesson 3-3

Similar Triangles: AA Similarity

Identify similar triangles using the AA Similarity Postulate

تحديد المثلثات المتشابهة باستخدام مسلّمة التشابه AA

Textbook · Exercises 1–8 · p. 133

KEY IDEA

AA Similarity: if two angles of one triangle are congruent to two angles of another, the triangles are similar. Shadows at the same time of day make similar triangles: **use proportions**.

التشابه AA: إذا طابقت زاويتان في مثلث زاويتين في مثلث آخر فالمثلثان متشابهان. الظلال في الوقت نفسه تكوّن مثلثات متشابهة فنحلّ بالتناسب.

MUST KNOW

- Two pairs of equal angles are enough.
- Find the third angle with $180°$ to compare.
- Shadow problems: object height ÷ shadow is the same.
- Write corresponding sides in the same order.

METHOD

1. Find the angles (use $180°$ for the third).
2. Decide: two pairs equal → similar.
3. Write a proportion.
4. Cross-multiply and solve.

WORKED EXAMPLE

Ex. 7 (P133): tower shadow 100 ft, Lia 4.5 ft with shadow $3\frac{1}{3}$ ft.

1. $\dfrac{h}{100} = \dfrac{4.5}{3.\overline{3}}$.
2. $h = 100 \cdot 1.35$.
3. $h = 135$ ft.

WATCH OUT

- Needing three pairs of equal angles.
- Mixing feet and inches.
- Writing the proportion with sides in different orders.
- Taking the distance to the lighthouse as the lighthouse's whole shadow.

CHECK YOUR ANSWER

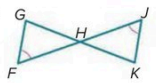
✓ The units match in the proportion.

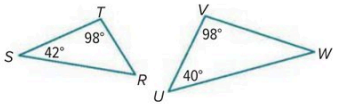
✓ Matching parts are in the same position.

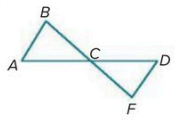
FROM YOUR TEXTBOOK Exercises 1–8 · p. 133

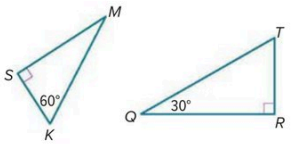
Example 1

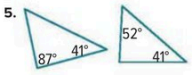
Determine whether each pair of triangles is similar. Explain your reasoning.

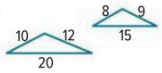
1. 

2. 

3. 

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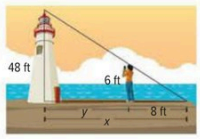
Example 2

7. CELL TOWERS A cell phone tower casts a shadow that is 100 feet long. At the same time, Lia stands near the tower and casts a shadow that is 3 feet 4 inches long. If Lia is 4 feet 6 inches tall, how tall is the cell phone tower?

8. LIGHTHOUSE Maya wants to know how far she is standing from a lighthouse. The end of Maya's shadow coincides with the end of the lighthouse's shadow.

a. What is the distance from the lighthouse to the end of the lighthouse's shadow, x?

b. What is the distance from Maya to the lighthouse, y?



Practice

Choose the correct answer. Show your working in the space beside each question.

9.1 **Are the triangles similar?**

$\triangle ABC$ has angles $60°$ and $30°$. $\triangle PQR$ has angles $30°$ and $90°$.

- (A) No: the triangles are not the same size
- (B) Yes, by SSS similarity
- (C) Not enough information
- (D) Yes, by AA similarity

9.2 **Are the triangles similar?**

One triangle has angles $41°$ and $87°$; another has angles $41°$ and $52°$.

- (A) No: only one pair of angles is equal
- (B) Yes, by AA similarity
- (C) No: $87° \neq 52°$
- (D) Not enough information

9.3 **Are the triangles similar?**

One triangle has angles $98°$ and $42°$; another has angles $98°$ and $40°$.

- (A) Yes, by AA similarity
- (B) Yes: both have $98°$
- (C) Not enough information
- (D) No: the second angles differ and so do the third angles

9.4 **Find the height.**

A cell tower casts a 100-ft shadow. At the same time Lia, 4 ft 6 in tall, casts a shadow of 3 ft 4 in. How tall is the tower?

- (A) 135 ft
- (B) 148 ft
- (C) 75 ft
- (D) 222 ft

9.5 **Find the distance.**

Maya is 6 ft tall and the lighthouse is 48 ft tall. The ends of their shadows coincide, and Maya's shadow is 8 ft long. How far is Maya from the lighthouse?

- (A) 64 ft
- (B) 8 ft
- (C) 56 ft
- (D) 48 ft

9.6 **Find the height.**

A 5-ft person casts a 4-ft shadow. At the same time a flagpole casts a 24-ft shadow. How tall is the flagpole?

- (A) 19.2 ft
- (B) 20 ft
- (C) 30 ft
- (D) 120 ft

9.7 **Which postulate proves similarity?**

Two angles of one triangle are congruent to two angles of another triangle.

- (A) Angle-Angle (AA) Similarity
- (B) Side-Side-Side (SSS) Congruence
- (C) Angle-Side-Angle (ASA) Congruence
- (D) Side-Angle-Side (SAS) Similarity

9.8 **Are the triangles similar?**

Right $\triangle ABC$ (right angle at B) has $m\angle A = 35°$. Right $\triangle DEF$ (right angle at E) has $m\angle D = 35°$.

- (A) No: they are different triangles
- (B) No: the hypotenuses may differ
- (C) Not enough information
- (D) Yes, by AA similarity

Section 10 | Coverage Q23 | Unit 3 · Lesson 3-4

Similar Triangles: SSS and SAS Similarity

Use similar triangles to solve problems

استخدام المثلثات المتشابهة في حلّ المسائل

Textbook · Exercises 5–11 · p. 139

KEY IDEA

Prove similarity with **AA**, **SSS** (all three ratios equal) or **SAS** (two ratios equal and the included angle equal). Then write a proportion with corresponding sides and solve.

أثبت التشابه بـ **AA** أو **SSS** (النسب الثلاث متساوية) أو **SAS** (نسبتان متساويتان والزاوية المحصورة متطابقة). ثم اكتب تناسبًا بالأضلاع المتناظرة وحلّه.

MUST KNOW

- SSS~: three equal ratios.
- SAS~: two equal ratios with the included angle.
- AA~: two equal angles.
- Same units in the proportion.

METHOD

1. Name the similar triangles and the reason.
2. Write corresponding sides as a proportion.
3. Cross-multiply and solve.
4. Answer with units.

WORKED EXAMPLE

Ex. 10 (P139): shadows, 6 ft person $\leftrightarrow$ 4 ft; tree $\leftrightarrow$ 18 ft.

1. $\frac{h}{18} = \frac{6}{4}$.
2. $h = 27$ ft.

WATCH OUT

- Writing sides in non-corresponding order.
- Mixing inches and feet.
- Using SAS with an angle that is not included.
- Forgetting to check that all three ratios are equal for SSS.

CHECK YOUR ANSWER

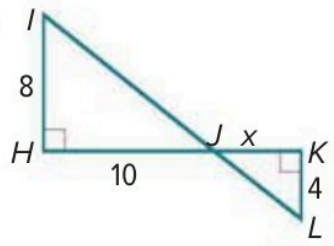
✓ The unknown side is in the right proportion position.

✓ The answer is reasonable (bigger triangle, bigger side).

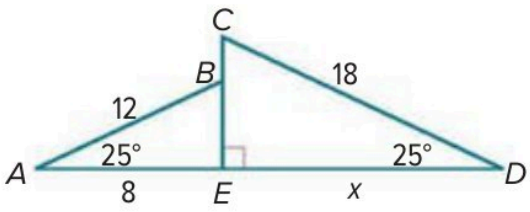
Example 2

Identify the similar triangles. Then find the value of *x*.

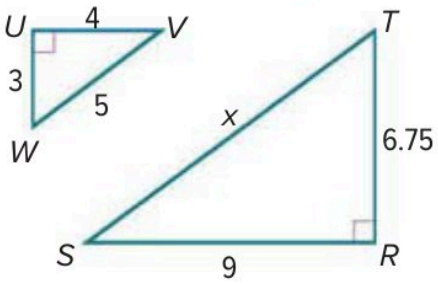
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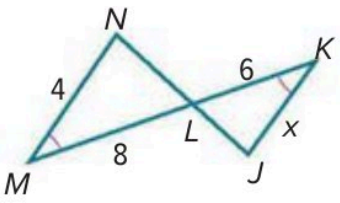
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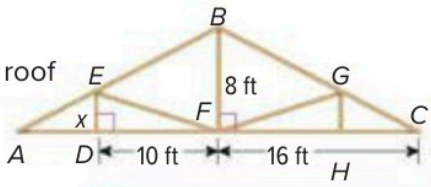


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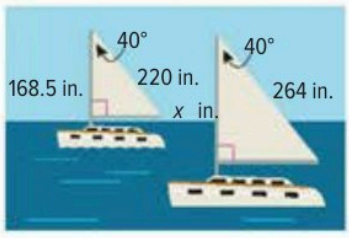
Example 3

9. ROOFING The skeleton of a roof is shown. Find the value of *x* such that triangles *DEF* and *FBC* in the outline of the roof are similar



10. RADIO A radio tower casts an 8-foot-long shadow at the same time that a vertical yardstick casts a shadow one half inch long. If the triangles formed by the objects and their shadows are similar, how tall is the radio tower?

11. SAILING The two sailboats shown are participating in a regatta. If the sails are similar, what is the value of *x*?



Practice

Choose the correct answer. Show your working in the space beside each question.

10.1 **Are the triangles similar?**

Triangle 1 has sides 6, 8, 10; triangle 2 has sides 9, 12, 15.

(A) No: the sides are different

(B) Yes, by SSS similarity (scale factor 1.5)

(C) Yes, by SAS similarity

(D) Not enough information

10.2 **Are the triangles similar?**

In $\triangle ABC$: $AB = 6$, $AC = 9$, $m\angle A = 70°$. In $\triangle DEF$: $DE = 10$, $DF = 15$, $m\angle D = 70°$.

(A) No: the included angle is not given for both

(B) Yes, by AA similarity

(C) No: the sides are not equal

(D) Yes, by SAS similarity

10.3 **Find x.**

Triangles with sides $6, 8, x$ and $9, 12, 15$ are similar.

(A) $x = 7.5$

(B) $x = 10$

(C) $x = 12$

(D) $x = 22.5$

10.4 **Find the height.**

A 6-ft person casts a 4-ft shadow. A tree casts an 18-ft shadow. How tall is the tree?

(A) 12 ft

(B) 48 ft

(C) 108 ft

(D) 27 ft

10.5 **Find the height.**

A radio tower casts an 8-ft shadow when a vertical yardstick (36 in) casts a $\frac{1}{2}$-in shadow. How tall is the tower?

(A) 576 ft

(B) 288 ft

(C) 48 ft

(D) 5760 ft

10.6 **Find x.**

Two sails are similar. The first has sides 220 in and 168.5 in; the corresponding sides of the second are 264 in and x in. Round to the nearest tenth.

(A) $x = 140.4$

(B) $x = 159.5$

(C) $x = 202.2$

(D) $x = 48.5$

10.7 **Are the triangles similar?**

Triangle 1 has sides $5, 7, 9$; triangle 2 has sides $10, 14, 20$.

(A) Yes, by SSS similarity

(B) Yes, by SAS similarity

(C) No: the ratios $\frac{1}{2}, \frac{1}{2}, \frac{9}{20}$ are not all equal

(D) Yes, by AA similarity

10.8 **Find the length.**

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$ with D on $\overline{AB}$ and E on $\overline{AC}$. If $AD = 6$, $AB = 15$ and $DE = 8$, find BC.

(A) 20

(B) 3.2

(C) 12.5

(D) 5

Section 11 · Coverage Q13 · Unit 3 · Lesson 3-5

Triangle Proportionality

Use proportional parts within triangles

استخدام الأجزاء المتناسبة داخل المثلثات

Textbook · Exercises 11–16 · p. 146

KEY IDEA

Triangle Proportionality Theorem: a line parallel to one side divides the other two sides proportionally. Its **converse** also holds. A **midsegment** is parallel to the third side and **half as long**.

نظرية التناسب في المثلث: المستقيم الموازي لأحد الأضلاع يقسم الضلعين الآخرين بالتناسب. وعكسها صحيح.
القطعة الواصلة بين منتصفي ضلعين توازي الضلع الثالث وطولها **نصف طوله**.

MUST KNOW

- Parallel line $\Rightarrow \dfrac{AD}{DB} = \dfrac{AE}{EC}$.
- Midsegment $= \frac{1}{2}$ of the third side.
- To prove parallel, check the two ratios are equal.
- Parallel lines cut transversals proportionally.

METHOD

1. Mark which segment is parallel.
2. Write the proportion of matching parts.
3. Cross-multiply and solve.
4. For a midsegment, use half the third side.

WORKED EXAMPLE

$AD = 6,\ DB = 9,\ AE = 8$**: find** EC **if** $DE \parallel BC$**.**

1. $\dfrac{6}{9} = \dfrac{8}{EC}$.
2. $6 \cdot EC = 72$.
3. $EC = 12$.

WATCH OUT

- Comparing the part of one side to the **whole** of the other.
- Using the midsegment as equal to the third side.
- Writing the two ratios in different orientations.
- Calling a segment parallel without checking.

CHECK YOUR ANSWER

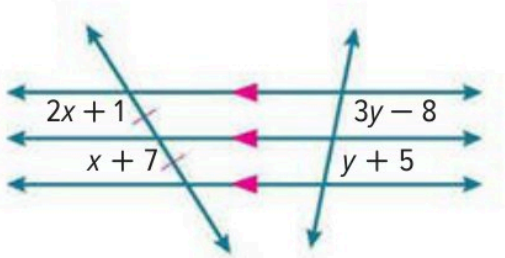
✓ Both ratios are written with the same orientation (top part over bottom part).

✓ A midsegment answer is half the side.

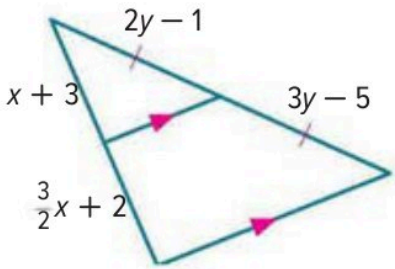
Example 5

Find the values of *x* and *y*.

11.



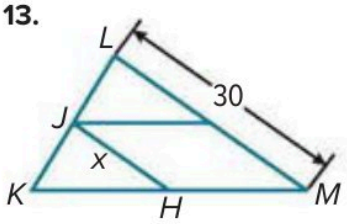
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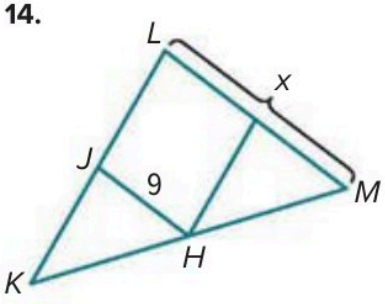
Mixed Exercises

JH **is a midsegment of △*KLM*. Find the value of *x*.**

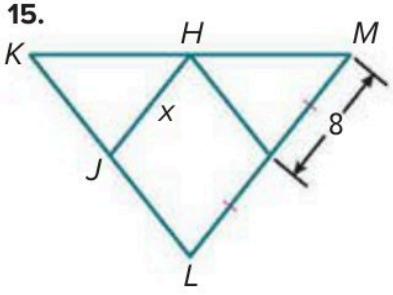
13.



14.



15.



16. In △*ABC*, $\overline{DE}$ is parallel to $\overline{AC}$ and $DE = 10$. Find the length of $\overline{AC}$ if $\overline{DE}$ is a midsegment of △*ABC*.



Practice

Choose the correct answer. Show your working in the space beside each question.

11.1 **Find the length.**

In $\triangle ABC$, D is on $\overline{AB}$ and E is on $\overline{AC}$ with $\overline{DE} \parallel \overline{BC}$. $AD = 6$, $DB = 9$, $AE = 8$. Find EC.

(A) 5.3 (B) 7.5

(C) 12 (D) 13.5

11.2 **Find x.**

$\overline{JH}$ is a midsegment of $\triangle KLM$ and it is parallel to $\overline{LM} = 30$. $JH = x$.

(A) $x = 60$ (B) $x = 15$

(C) $x = 30$ (D) $x = 10$

11.3 **Find x.**

$\overline{JH}$ is a midsegment of $\triangle KLM$. $JH = x + 4$ and the parallel side is $4x - 4$.

- (A) $x = 2$
- (B) $x = 8$
- (C) $x = 4$
- (D) $x = 6$

11.4 **Find the length.**

In $\triangle ABC$, $\overline{DE} \parallel \overline{AC}$ and $DE = 10$, where $\overline{DE}$ is a midsegment. Find AC.

- (A) 5
- (B) 20
- (C) 10
- (D) 40

11.5 **Find the length of the beam.**

An A-frame has a 24-ft base. A horizontal beam is fixed halfway up, parallel to the ground. How long is the beam?

- (A) 24 ft
- (B) 6 ft
- (C) 48 ft
- (D) 12 ft

11.6 **Is the segment parallel?**

D is on $\overline{AB}$ and E on $\overline{AC}$ with $AD = 4,\ DB = 6,\ AE = 6,\ EC = 9$. Is $\overline{DE} \parallel \overline{BC}$?

- (A) Yes: $\frac{4}{6} = \frac{6}{9}$
- (B) No: $4 \neq 6$
- (C) Yes: $4 + 6 = 6 + 9$
- (D) Not enough information

11.7 **Is the segment parallel?**

$AD = 3,\ DB = 5,\ AE = 4,\ EC = 6$. Is $\overline{DE} \parallel \overline{BC}$?

- (A) Yes: $\frac{3}{5} \approx \frac{4}{6}$
- (B) Yes: $3 + 5 = 4 + 6 - 2$
- (C) No: $\frac{3}{5} \neq \frac{4}{6}$
- (D) Not enough information

11.8 **Find x.**

Three parallel lines cut two transversals. On the first transversal the segments are 4 and 6; on the second they are 6 and x.

- (A) $x = 4$
- (B) $x = 8$
- (C) $x = 9$
- (D) $x = 12$

Section 12 | Coverage Q14 | Unit 3 · Lesson 3-6

Parts of Similar Triangles

Recognize and use proportional relationships of corresponding angle bisectors, altitudes, and medians of similar triangles

التعرّف إلى العلاقات المتناسبة بين منصّفات الزوايا والارتفاعات والمتوسطات المتناظرة في المثلثات المتشابهة واستخدامها

Textbook · Exercises 1–11 · p. 153

KEY IDEA

In similar triangles the **altitudes**, **medians** and **angle bisectors** from corresponding vertices are in the **same ratio** as the corresponding sides (the scale factor). Perimeters also follow it.

في المثلثات المتشابهة تكون **الارتفاعات** و**المتوسطات** و**منصّفات الزوايا** المتناظرة بنفس **نسبة الأضلاع** المتناظرة (معامل التشابه). والمحيطات كذلك.

MUST KNOW

- Scale factor = ratio of any pair of corresponding sides.
- The same ratio holds for altitudes, medians, bisectors and perimeters.
- The ratio of areas is the square of the scale factor.
- Match the correct corresponding parts.

METHOD

1. Find the scale factor from the given sides.
2. Write the proportion for the required segment.
3. Cross-multiply and solve.
4. Check the larger triangle has the larger answer.

WORKED EXAMPLE

Ex. 5 (P153): $ST = 6,\ SH = 5,\ FJ = 7$**. Find** FG**.**

1. $\frac{ST}{FG} = \frac{SH}{FJ}$.
2. $\frac{6}{FG} = \frac{5}{7}$.
3. $FG = 8.4$.

WATCH OUT

- Putting the two triangles in different orders in the ratio.
- Using the area ratio for lengths.
- Mixing a median with an altitude.
- Forgetting to find the scale factor first.

CHECK YOUR ANSWER

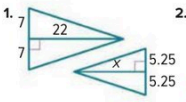
✓ The unknown belongs to the right triangle in the ratio.

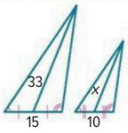
✓ The bigger triangle's segment is bigger.

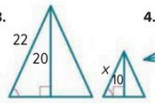
FROM YOUR TEXTBOOK Exercises 1–11 · p. 153

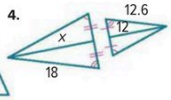
Example 1

Each pair of triangles is similar. Find the value of *x*.

1. 

2. 

3. 

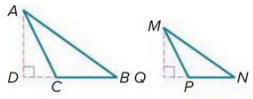
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Example 2

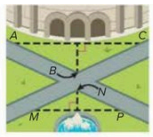
5. If $\triangle RST \sim \triangle EFG$, $\overline{SH}$ is an altitude of $\triangle RST$, $\overline{FJ}$ is an altitude of $\triangle EFG$, $ST = 6$, $SH = 5$, and $FJ = 7$, find FG.



6. If $\triangle ABC \sim \triangle MNP$, $\overline{AD}$ is an altitude of $\triangle ABC$, $\overline{MQ}$ is an altitude of $\triangle MNP$, $AB = 24$, $AD = 14$, and $MQ = 10.5$, find MN.

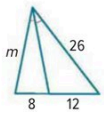


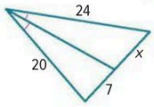
7. SCENERY The intersection of the two paths shown forms two similar triangles. If AC is 50 yards, MP is 35 yards, and the fountain is 5 yards from the intersection, about how far from the intersection is the stadium entrance? Round to the nearest yard.



Example 3

Find the value of each variable to the nearest tenth.

8. 

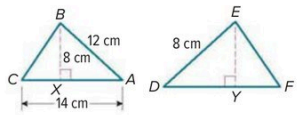
9. 

Mixed Exercises

For Exercises 10 and 11, $\triangle ABC \sim \triangle DEF$.

10. Find the length of $\overline{XC}$ to the nearest tenth.

11. Find the length of $\overline{EY}$ to the nearest tenth.



Practice

Choose the correct answer. Show your working in the space beside each question.

12.1 **Find the length.**

$\triangle RST \sim \triangle EFG$. $\overline{SH}$ is an altitude of $\triangle RST$ and $\overline{FJ}$ is the matching altitude of $\triangle EFG$. $ST = 6$, $SH = 5$, $FJ = 7$. Find FG.

(A) 8.4 (B) 4.3 (C) 7 (D) 5.8

12.2 **Find the length.**

$\triangle ABC \sim \triangle MNP$. $\overline{AD}$ and $\overline{MQ}$ are corresponding altitudes. $AB = 24$, $AD = 14$, $MQ = 10.5$. Find MN.

(A) 32 (B) 13.5 (C) 18 (D) 24.5

12.3 **Find the median.**

Two similar triangles have corresponding sides in the ratio $3 : 5$. The median of the smaller triangle is 9. The corresponding median of the larger is

(A) 5.4 (B) 15 (C) 12 (D) 27

12.4 **Find the angle bisector.**

$\triangle ABC \sim \triangle DEF$ with scale factor $2 : 3$. The angle bisector from A is 8. The corresponding bisector in $\triangle DEF$ is

(A) 5.3 (B) 10 (C) 16 (D) 12

12.5 **Find the altitude.**

Two similar triangles have scale factor $4:7$. The altitude of the larger triangle is 21. The matching altitude of the smaller triangle is

(A) 36.75

(B) 12

(C) 9

(D) 15

12.6 **Find the altitude.**

Two similar triangles have perimeters 20 and 30. The altitude of the smaller is 8. The matching altitude of the larger is

(A) 5.3

(B) 10

(C) 18

(D) 12

12.7 **Find x.**

Two similar triangles have corresponding sides 12 and 18. Their corresponding altitudes are x and 15. Find x.

(A) $x = 10$

(B) $x = 22.5$

(C) $x = 9$

(D) $x = 6$

12.8 **Find the angle bisector.**

Two similar triangles have corresponding sides 24 and 36. The bisector of the smaller one's angle is 10. What is the corresponding bisector of the larger?

(A) 6.7

(B) 12

(C) 15

(D) 26

Answer key

Each answer with the reason in one line.

Section 1 · Coverage Q5 · Lesson 2-1

1.1	**B**	$(n-2)\cdot 180^\circ = (6-2)\cdot 180^\circ = 720^\circ$.
1.2	**A**	Sum $= (8-2)\cdot 180^\circ = 1080^\circ$; each angle $= 1080^\circ \div 8 = 135^\circ$.
1.3	**C**	The sum is $(4-2)\cdot 180^\circ = 360^\circ$: $6x+12 = 360 \Rightarrow x = 58$. The angles are $121^\circ, 58^\circ, 58^\circ, 123^\circ$; the largest is $m\angle S = 123^\circ$.
1.4	**B**	The sum is $(5-2)\cdot 180^\circ = 540^\circ$: $90+90+(2x-20)+x+(2x+10) = 540 \Rightarrow 5x+170 = 540 \Rightarrow x = 74$. So $m\angle E = 2(74)+10 = 158^\circ$.
1.5	**D**	Each exterior angle is $180^\circ - 144^\circ = 36^\circ$, so $n = 360^\circ \div 36^\circ = 10$. (Or solve $\frac{(n-2)180}{n} = 144$.)
1.6	**B**	Exterior angle $= 20^\circ$, so $n = 360 \div 20 = 18$.
1.7	**D**	Sum $= (7-2)\cdot 180^\circ = 900^\circ$; $900 \div 7 \approx 128.6^\circ$.
1.8	**A**	Sum $= 10\cdot 180^\circ = 1800^\circ$; each angle $= 1800 \div 12 = 150^\circ$.

Section 2 · Coverage Q22 · Lesson 2-1

2.1	**D**	The exterior angles of **any** convex polygon add up to 360°, whatever the number of sides.
2.2	**B**	$360^\circ \div 10 = 36^\circ$.
2.3	**A**	$n = 360^\circ \div 24^\circ = 15$. (The matching interior angle is 156°.)
2.4	**C**	$n = 360 \div 40 = 9$.
2.5	**B**	They add up to 360°: $8x+8 = 360 \Rightarrow x = 44$. The angles are $134^\circ, 88^\circ, 89^\circ, 49^\circ$.
2.6	**D**	Interior $= 180^\circ - 18^\circ = 162^\circ$ (the polygon has $360 \div 18 = 20$ sides).
2.7	**B**	Exterior angle $= 180^\circ - 108^\circ = 72^\circ$; $n = 360 \div 72 = 5$.
2.8	**D**	They add up to 360°: $4x-12 = 360 \Rightarrow x = 93$. The angles are $82^\circ, 103^\circ, 31^\circ, 144^\circ$.

Section 3 · Coverage Q6 · Lesson 2-2

3.1	**A**	Consecutive angles of a parallelogram are supplementary: $m\angle S = 180^\circ - 128^\circ = 52^\circ$. (The opposite angle R is 128°.)
3.2	**D**	Opposite sides of a parallelogram are congruent: $SR = PQ = 3$ and $SP = QR = 5$.
3.3	**B**	Opposite sides are equal: $3b+5 = b+11 \Rightarrow 2b = 6 \Rightarrow b = 3$. Each of these sides is 14. (Likewise $a+15 = 3a+11 \Rightarrow a = 2$.)
3.4	**A**	Consecutive angles are supplementary: $(x-5)+(2x+11) = 180 \Rightarrow 3x+6 = 180 \Rightarrow x = 58$. The angles are 53° and 127°.
3.5	**C**	The diagonals bisect each other, so $JP = PL$: $3y-5 = y+5 \Rightarrow y = 5$. Then $JP = PL = 10$ and $JL = 20$.
3.6	**B**	Take the vectors from $(-8, 0)$: $u = (4, -5)$ and $v = (13, 5)$. Area $= \lvert 4\cdot 5 - (-5)\cdot 13\rvert = 85$ square units. Each unit is 5 m, so $85 \cdot 25 = 2125$ m².
3.7	**D**	Congruent diagonals belong to rectangles, not to every parallelogram.
3.8	**B**	Opposite angles are equal and consecutive angles are supplementary: $70, 110, 70, 110$.

Section 4 · Coverage Q7 · Lesson 2-3

4.1	**C**	Theorem 2.9: if both pairs of opposite sides are congruent, the quadrilateral is a parallelogram.
4.2	**A**	Theorem 2.10: if both pairs of opposite angles are congruent, it is a parallelogram.
4.3	**D**	Theorem 2.11: if the diagonals bisect each other, the quadrilateral is a parallelogram.
4.4	**B**	Theorem 2.12 needs one pair of opposite sides that is **both parallel and congruent**. Congruent alone is not enough (an isosceles trapezoid has equal legs).
4.5	**A**	Opposite sides are equal: $6x+1 = 8x-3 \Rightarrow x = 2$ and $4y-7 = 2y+3 \Rightarrow y = 5$.
4.6	**C**	Opposite angles are equal: $5x+29 = 7x-11 \Rightarrow x = 20$; $5y-9 = 3y+15 \Rightarrow y = 12$.
4.7	**B**	The diagonals must bisect each other: $-4x-2 = -3x+4 \Rightarrow x = -6$ and $3y-5 = 2y+18 \Rightarrow y = 23$. Each half then measures 22 and 64.
4.8	**D**	Both pairs of opposite sides are parallel and congruent (equal vectors), so the quadrilateral is a parallelogram.

Section 5 · Coverage Q8 · Lesson 2-6

5.1 **C** $\triangle BEC$ is a right triangle: $BC = \sqrt{4^2 + 7^2} = \sqrt{65}$, which cannot be simplified. ($AB = \sqrt{11^2 + 7^2} = \sqrt{170}$.)

5.2 **C** Perimeter $= 2(\sqrt{170} + \sqrt{65}) \approx 2(13.038 + 8.062) \approx 42.2$.

5.3 **A** The four angles add up to $360°$: $2m\angle Q = 360 - 110 - 48 = 202$, so $m\angle Q = 101°$.

5.4 **D** $2m\angle D = 360° - 36° - 70° = 254°$, so $m\angle D = 127°$.

5.5 **B** The diagonals of a kite are perpendicular (the one on the axis of symmetry bisects the other).

5.6 **A** The diagonals of an isosceles trapezoid are congruent: $3x - 7 = 2x + 8 \Rightarrow x = 15$ (each is 38).

5.7 **C** The base angles of an isosceles trapezoid are congruent: $4x + 11 = 2x + 33 \Rightarrow x = 11$ (each is $55°$).

5.8 **B** Midsegment $= \frac{1}{2}(b_1 + b_2) = \frac{1}{2}(11 + 18) = 14.5$.

Section 6 · Coverage Q9 · Lesson 3-1

6.1 **A** A dilation about the origin multiplies both coordinates by k: $(-10 \cdot 0.25,\ 4 \cdot 0.25) = (-2.5,\ 1)$.

6.2 **C** $(2 \cdot \frac{5}{4},\ -1 \cdot \frac{5}{4}) = (\frac{5}{2},\ -\frac{5}{4})$. Dividing by $\frac{5}{4}$ instead of multiplying gives the wrong answer $(\frac{8}{5}, -\frac{4}{5})$.

6.3 **C** $(0 \cdot \frac{3}{4},\ -8 \cdot \frac{3}{4}) = (0,\ -6)$. The image stays on the y-axis. (Dividing by $\frac{3}{4}$ gives the wrong $(0, -10\frac{2}{3})$.)

6.4 **A** $(-2 \cdot 1.5,\ 6 \cdot 1.5) = (-3,\ 9)$.

6.5 **D** $(5 \cdot 2.5,\ 2 \cdot 2.5) = (12.5,\ 5)$.

6.6 **B** $0 < k < 1$ gives a reduction; $k > 1$ gives an enlargement; $k = 1$ gives a congruent image.

6.7 **A** $k = \frac{A'}{A} = \frac{6}{2} = \frac{9}{3} = 3$ (an enlargement).

6.8 **C** $k = \frac{\text{image length}}{\text{original length}} = \frac{10}{4} = 2.5$.

Section 7 · Coverage Q11 · Lesson 3-1

7.1 **D** $\frac{1}{3} = \frac{2}{6} = \frac{-2}{-6} = \frac{1}{3}$. The image is smaller, so it is a reduction.

7.2 **A** $\frac{5}{2} = \frac{10}{4} = \frac{15}{6} = \frac{5}{2} = 2.5$.

7.3 **C** $k = \frac{15}{6} = 2.5$.

7.4 **C** $k = \frac{A'B'}{AB} = \frac{6}{8} = \frac{3}{4}$: a reduction.

7.5 **A** The image is on the opposite side of the center: the scale factor is negative, $k = -3$.

7.6 **D** Scale factors multiply: $2 \cdot \frac{1}{3} = \frac{2}{3}$.

7.7 **B** $k \cdot \frac{1}{k} = 1$: a scale factor of 1 leaves the figure unchanged.

7.8 **A** Since $0 < \frac{5}{8} < 1$, the image is smaller: a reduction.

Section 8 · Coverage Q10 · Lesson 3-2

8.1 **B** Corresponding sides are proportional: $\frac{8}{6} = \frac{12}{x} \Rightarrow x = \frac{6 \cdot 12}{8} = 9$.

8.2 **D** $\frac{x+3}{15} = \frac{4}{10} \Rightarrow x + 3 = 6 \Rightarrow x = 3$.

8.3 **A** $\frac{2x+2}{15} = \frac{4}{10} \Rightarrow 2x + 2 = 6 \Rightarrow x = 2$.

8.4 **C** Length $= 5k$, width $= 2k$: $2(5k + 2k) = 126 \Rightarrow k = 9$, so the width is $2 \cdot 9 = 18$ m (and the length is 45 m).

8.5 **C** 8 cm $\leftrightarrow$ 4 km, so 1 cm $= 0.5$ km. The map perimeter is $2 + 8 + 7 = 17$ cm, and $17 \cdot 0.5 = 8.5$ km.

8.6 **A** $\frac{6}{8} = \frac{9}{12} = \frac{3}{4}$, and all angles are $90°$. The rectangles are similar; from the larger to the smaller the factor is $\frac{3}{4}$ (from the smaller to the larger it is $\frac{4}{3}$).

8.7 **D** $\frac{4}{6} = \frac{2}{3}$ but $\frac{6}{8} = \frac{3}{4}$. Similar polygons need **both** congruent angles **and** proportional sides.

8.8 **B** Corresponding angles of similar polygons are congruent: $\angle X$ corresponds to $\angle B$.

Section 9 · Coverage Q12 · Lesson 3-3

9.1 **D** $\triangle ABC$ has angles $60°, 30°, 90°$ and $\triangle PQR$ has $30°, 90°, 60°$: two pairs of equal angles, so they are similar by AA.

9.2 **B** The third angles are $180 - 41 - 87 = 52°$ and $180 - 41 - 52 = 87°$, so the angles are $41, 87, 52$ in both triangles. They are similar.

9.3 **D** The third angles are $40°$ and $42°$: only one pair of equal angles, so AA fails.

9.4 **A** The sun makes the same angle for both, so the triangles are similar: $\frac{h}{100} = \frac{4.5}{3.3} \Rightarrow h = 135$ ft.

9.5 **C** The lighthouse's shadow is x: $\frac{48}{6} = \frac{x}{8} \Rightarrow x = 64$ ft. Maya stands $64 - 8 = 56$ ft from the lighthouse.

9.6 **C** $\frac{h}{24} = \frac{5}{4} \Rightarrow h = 30$ ft.

9.7 **A** AA Similarity needs only two pairs of congruent angles: the third pair follows.

9.8 **D** Each has a right angle and a $35°$ angle: two pairs of congruent angles, so AA applies.

Section 10 · Coverage Q23 · Lesson 3-4

10.1 **B** $\frac{6}{9} = \frac{8}{12} = \frac{10}{15} = \frac{2}{3}$: all three pairs of sides are proportional.

10.2 **D** $\frac{6}{10} = \frac{9}{15} = \frac{3}{5}$ and the included angles are equal (70°): SAS similarity.

10.3 **B** $\frac{6}{9} = \frac{x}{15} \Rightarrow x = \frac{6 \cdot 15}{9} = 10$.

10.4 **D** Similar triangles: $\frac{h}{18} = \frac{6}{4} \Rightarrow h = 27$ ft.

10.5 **A** Work in inches: the tower's shadow is 96 in. $\frac{h}{96} = \frac{36}{0.5} \Rightarrow h = 6912$ in $= 576$ ft.

10.6 **C** $\frac{220}{264} = \frac{168.5}{x} \Rightarrow x = \frac{168.5 \cdot 264}{220} \approx 202.2$.

10.7 **C** $\frac{5}{10} = \frac{7}{14} = \frac{1}{2}$ but $\frac{9}{20} \neq \frac{1}{2}$: not similar.

10.8 **A** $\triangle ADE \sim \triangle ABC$ (AA): $\frac{6}{15} = \frac{8}{BC} \Rightarrow BC = 20$.

Section 11 · Coverage Q13 · Lesson 3-5

11.1 **C** Triangle Proportionality Theorem: $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{6}{9} = \frac{8}{EC} \Rightarrow EC = 12$.

11.2 **B** A midsegment is half as long as the side it is parallel to: $x = \frac{1}{2} \cdot 30 = 15$.

11.3 **D** $x + 4 = \frac{1}{2}(4x - 4) \Rightarrow 2x + 8 = 4x - 4 \Rightarrow x = 6$: the midsegment is 10 and the side is 20.

11.4 **B** $DE = \frac{1}{2}AC$, so $AC = 2 \cdot 10 = 20$.

11.5 **D** The beam is a midsegment (it joins the midpoints of the two sides): half of 24 ft $= 12$ ft.

11.6 **A** Converse of the Triangle Proportionality Theorem: if $\frac{AD}{DB} = \frac{AE}{EC}$, the segment is parallel to the third side. Here both ratios equal $\frac{2}{3}$.

11.7 **C** $\frac{3}{5} = 0.6$ but $\frac{4}{6} = 0.\overline{6}$. The sides are not divided proportionally, so $\overline{DE}$ is not parallel to $\overline{BC}$.

11.8 **C** Parallel lines cut transversals proportionally: $\frac{4}{6} = \frac{6}{x} \Rightarrow x = 9$.

Section 12 · Coverage Q14 · Lesson 3-6

12.1 **A** Corresponding altitudes are proportional to corresponding sides: $\frac{ST}{FG} = \frac{SH}{FJ} \Rightarrow \frac{6}{FG} = \frac{5}{7} \Rightarrow FG = 8.4$.

12.2 **C** $\frac{AB}{MN} = \frac{AD}{MQ} \Rightarrow \frac{24}{MN} = \frac{14}{10.5} = \frac{4}{3} \Rightarrow MN = 18$.

12.3 **B** Corresponding medians have the same ratio as the sides: $9 \cdot \frac{5}{3} = 15$.

12.4 **D** $8 \cdot \frac{3}{2} = 12$.

12.5 **B** $21 \cdot \frac{4}{7} = 12$.

12.6 **D** The ratio of perimeters equals the scale factor $20 : 30 = 2 : 3$, so the larger altitude is $8 \cdot \frac{3}{2} = 12$.

12.7 **A** $\frac{12}{18} = \frac{x}{15} \Rightarrow x = \frac{12 \cdot 15}{18} = 10$.

12.8 **C** Scale factor $36 : 24 = 3 : 2$: $10 \cdot \frac{3}{2} = 15$.